

# Thomas' Calculus

## Thirteen Edition.

Ex Find the gradient of the function at the given point

i)  $f(x, y) = y - x, (2, 1)$

ii)  $f(x, y) = \ln(x^2 + y^2), (1, 1)$

iii)  $f(x, y) = \tan^{-1} \sqrt{x/y}, (4, -2)$

Ex Find  $\nabla f$  at the given point

$f(x, y, z) = x^2 + y^2 + z^2 + z \ln x, (2, -1, 1)$

$f(x, y, z) = \sin^{-1} x + \tan^{-1} y/x + \ln z^2, (0, 1, 1)$

$f(x, y, z) = x^2 \ln(x^2 + y^2) + z^2 \sqrt{x^2 + y^2}, (1, 1, 1)$

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